Raffles Institution
Physics Department

WS07: Determining the Height of a Building **Time: 1 h 20 m**

Name: .... ( )

Class: 26S0 **Date:** ....

Instructions before Lesson

To complete the activities in this worksheet, you must download a copy of the Excel raw data file and rename it as "your_name.xlsx".

Introduction

We will determine the height of Block A using the acceleration and pressure data of the elevator collected by Phyphox.

(a) **(i)** Open the Excel file in Microsoft Excel.

(ii) Insert a *Scatter Plot* of acceleration *a* against time *t*.

From your graph, determine the times t_1 and t_2 when the acceleration starts and when the deceleration ends, respectively.

Retain 1 second of data before the acceleration starts and 1 second of data after the deceleration ends. Delete the rest of the data.

t_1 = 9.09 s

t_2 = 30.79 s

(b) **(i)** An elevator, moving with an initial velocity of *u*, experiences an average acceleration of $\langle a \rangle$ for a short amount of time Δt.

Write an expression for the final velocity *v* of the elevator.

Final velocity of elevator

$$v = u + \langle a \rangle \Delta t$$

(ii) In cell C1, enter the heading for velocity.

In cell C2, enter the value zero as the lift is assumed to be at rest.

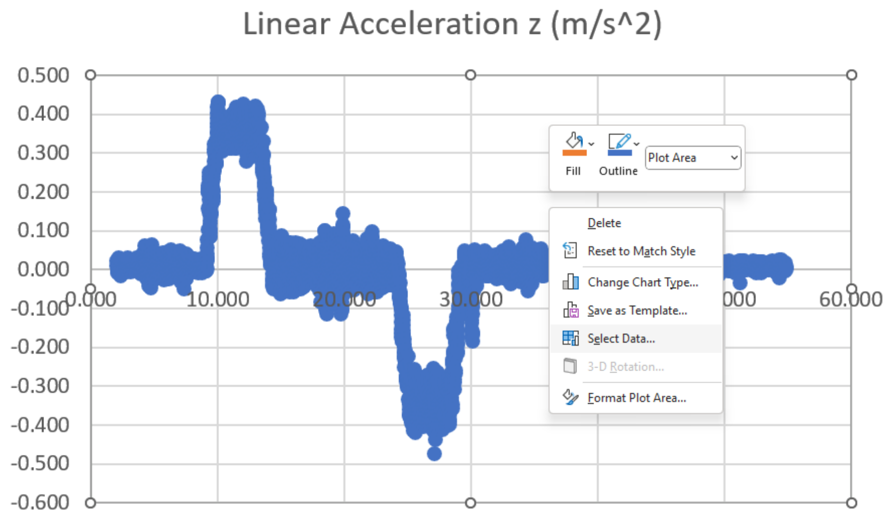
In cell C3, enter the formula for the final velocity.

"=C2+0.5*(B2+B3)*(A3-A2)" or "=C2+0.5*(B2+B3)*0.01"

(iii) Apply the formula to the entire column using the *fill* function.

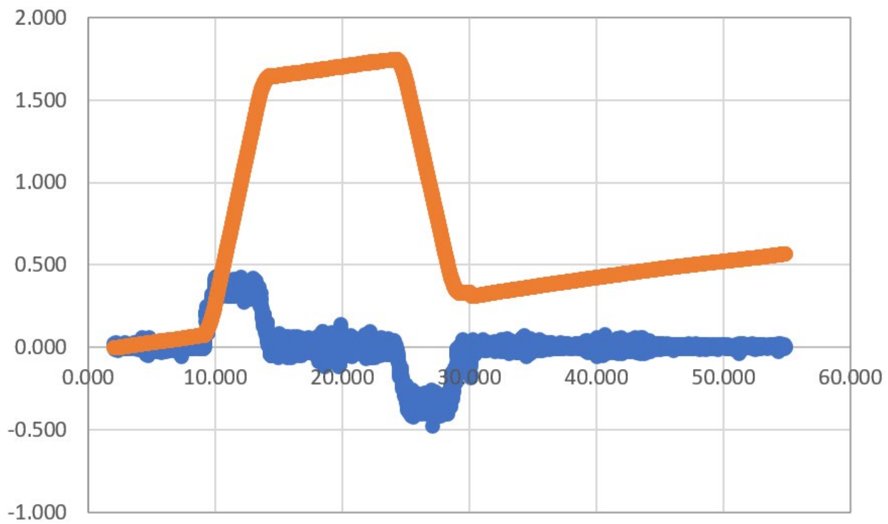
Select all data and set the decimal places of all cells to 3.

With the *Scatter Plot* selected, right-click on the *Plot Area*. A contextualised menu will appear. Click on *Select Data*.



In the *Legend Entries (Series)*, click on *Add* to insert another set of data.

Series name: Cell C1, Series X values: data in column 1, Series Y values: data in column 3.



(iv) A typical elevator will accelerate for a short time, then move with constant velocity and finally decelerate for a short time before coming to a stop at the desired level.

From the *v*-*t* graph, comment on whether the data agrees with the above description.

The elevator does not move with constant velocity when it is not accelerating.

The elevator's velocity is not zero when it is not moving.

(v) If the data does not agree with the description, suggest a reason why this is so.

Hence, suggest a method to correct the data so that the *v*-*t* graph will fit the description.

On inspection, there is non-zero acceleration (zero error) when the elevator is not accelerating.

The average non-zero acceleration (systematic error) should be subtracted from the acceleration data.

(c) **(i)** The lift is at rest at the start and end of its motion. This means that the average acceleration for the entire duration should be zero. Any non-zero average acceleration is taken to be the zero error in acceleration.

Determine the average acceleration $\langle a_T \rangle$ of the lift for the entire duration.

$\langle a_T \rangle$ = 0.0110 m s^{-2} ....

(ii) In column D, compute the corrected acceleration a_{cor} by applying the correction in **c(i)**.

In column E, compute the corrected velocity v_{cor} using the formula in **b(i)**.

Create a new scatter plot for the corrected acceleration a_{cor} and velocity v_{cor} against time *t*.

From your velocity-time graph, determine the velocity v_C of the elevator when it is travelling upwards with a constant velocity.

v_C = 1.51 m s^{-1} ....

(iii) Write an expression for the change in displacement Δs of the elevator, in terms of *u*, *v* and Δt .

Change in displacement of elevator: $\Delta s = \frac{1}{2}(u+v)\Delta t$

(iv) In cell F1, enter the heading for displacement.

In cell F2, enter the value zero as the lift is assumed to be at ground level.

In cell F3, enter the formula for the final displacement.

"=F2+0.5*(E2+E3)*(A3-A2)" or "=F2+0.5*(E2+E3)*0.01"

(v) Insert a *Scatter Plot* of displacement *s* against time *t*.

(vi) From your *s*-*t* graph, determine the height *H* of the elevator when it reaches the top floor of the building.

H = 22.8 m ....

(d) **(i)** Using the pressure data set, insert a scatter plot of pressure p against time t.

Using the graph, determine the average atmospheric pressure p_1 and p_2 at the ground floor and top floor, respectively.

Find the average pressure from the first 10 s. p_1 = 100764 Pa

Find the average pressure from the last 20 s. p_2 = 100504 Pa

(ii) Hence, by considering the air pressure difference between p_1 and p_2 and any other appropriate formulas, determine the height H' of the elevator when it travels from the ground floor to the top floor of the building.

Assume that the density of air is 1.204 kg m^{-3}.

Since $\Delta p = \rho g H'$,

$$H' = \frac{\Delta p}{\rho g} = \frac{100764 - 100504}{1.204 \times 9.81} = \frac{260}{11.8} = 22.0 \text{ m}$$

H' = 22.0 m

(iii) Determine the percentage difference between your calculated values of H and H'.

Comment on whether the values agree with each other, and explain if there are any significant differences between H and H'.

$$\% \text{ difference} = \frac{22.8 - 22.0}{22.8} \times 100\% = 3.5\%$$

or

$$\% \text{ difference} = \frac{22.8 - 22.0}{22.0} \times 100\% = 3.6\%$$

There is no significant difference between H and H' since the percentage difference between H and H' is less than 5%.

End of Lesson Reflection

(a) List down **THREE** concepts/ideas that I have learnt in this lesson.

1.

2.

3.

(b) List down some concepts/ideas that I still have doubts about.

(c) I would like to learn more about…

END OF WORKSHEET